Lab 7: Fuzzing - Screenshots (terminal output captured from real runs)

Step 2: python processor.py (thermostat_control.py, original)

```
$ python thermostat_control.py
==== Running parse_setpoint ====
Enter a setpoint (e.g. 21.5C or 70F): Setpoint in Celsius: 21.5

==== Running average_temperature ====
Enter sensor readings as CSV (e.g. 20.5,21,22.5): Average Temperature: 21.333333333333332

==== Running schedule_slot ====
Enter a time as HH:MM (e.g. 07:30): Schedule Slot: day 30
```

Step 4: pytest test_thermostat_control.py (original code) - failing

```
    |     parse_setpoint(s)
    |     assert False, f"parse_setpoint crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_setpoint crashed with: ValueError("could not convert string to float: ''") on input ''
    |     s='',
    |     parse_setpoint(s)
    |     assert False, f"parse_setpoint crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_setpoint crashed with: TypeError("'NoneType' object is not subscriptable") on input None
    |     s=None,  # or any other generated value
    |     assert False, f"average_temperature crashed with: {e!r} on input {s!r}"
    | AssertionError: average_temperature crashed with: ValueError("could not convert string to float: ''") on input ''
    |     s='',
    |     assert False, f"average_temperature crashed with: {e!r} on input {s!r}"
    | AssertionError: average_temperature crashed with: AttributeError("'NoneType' object has no attribute 'split'") on inp
    |     s=None,
    |     assert False, f"schedule_slot crashed with: {e!r} on input {s!r}"
    | AssertionError: schedule_slot crashed with: ValueError("invalid literal for int() with base 10: ''") on input ':'
    |     s=':',
    |     assert False, f"schedule_slot crashed with: {e!r} on input {s!r}"
    | AssertionError: schedule_slot crashed with: ValueError('not enough values to unpack (expected 2, got 1)') on input ''
    |     s='',  # or any other generated value
    |     assert False, f"schedule_slot crashed with: {e!r} on input {s!r}"
    | AssertionError: schedule_slot crashed with: AttributeError("'NoneType' object has no attribute 'split'") on input Non
    |     s=None,  # or any other generated value
FAILED test_thermostat_control.py::test_parse_setpoint_no_crash - ExceptionGr...
FAILED test_thermostat_control.py::test_average_temperature_no_crash - Except...
FAILED test_thermostat_control.py::test_schedule_slot_no_crash - ExceptionGro...
============================= 3 failed in 18.77s ==============================
```

Step 5: python thermostat_control.py (fixed)

```
==== Running parse_setpoint ====
Enter a setpoint (e.g. 21.5C or 70F): Setpoint in Celsius: 21.5

==== Running average_temperature ====
Enter sensor readings as CSV (e.g. 20.5,21,22.5): Average Temperature: 21.333333333333332

==== Running schedule_slot ====
Enter a time as HH:MM (e.g. 07:30): Schedule Slot: day 30
==== Running parse_setpoint ====
Enter a setpoint (e.g. 21.5C or 70F): Setpoint in Celsius: Invalid input

==== Running average_temperature ====
Enter sensor readings as CSV (e.g. 20.5,21,22.5): Average Temperature: Invalid input

==== Running schedule_slot ====
Enter a time as HH:MM (e.g. 07:30): Schedule Slot: Invalid input
```

Step 6: pytest test_thermostat_control.py (fixed code) - passing

```
============================= test session starts ==============================
platform win32 -- Python 3.12.10, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\adhithya\AppData\Local\Microsoft\WindowsApps\Pytho
hypothesis profile 'default'
rootdir: C:\Users\adhithya\AppData\Local\Temp\claude\d7c4855a9c206ba1\d9cc6970-0377-4ee1-af50-cf814dbee6af\scratchpad\SE_CS
plugins: anyio-4.12.0, hypothesis-6.170.15, cov-7.1.0
collecting ... collected 3 items

test_thermostat_control.py::test_parse_setpoint_no_crash PASSED          [ 33%]
test_thermostat_control.py::test_average_temperature_no_crash PASSED     [ 66%]
test_thermostat_control.py::test_schedule_slot_no_crash PASSED           [100%]

============================== 3 passed in 8.57s ===============================
```